PROJECT VARUNA (AERO-SENSE)

NEXT-GEN MULTI-HAZARD EARLY WARNING & EVACUATION COMMAND PLATFORM

Hackathon / Year:	Smart India Hackathon 2026 (SIH 2026)	**Problem Statement ID:**	SIH26078
Team Name:	Team AETHON	**Project Codename:**	VARUNA (AERO-SENSE)
Sponsoring Body:	Ministry of Earth Sciences (MoES) / NCMR	**Live Deployment URL:**	https://varuna-access.vercel.app
Access Credentials:	ID: OP-8842-MET | Password: IMD@2026	**Classification:**	Official Technical Documentation

1. EXECUTIVE SUMMARY & OPERATIONAL PARADIGM

Project VARUNA (AERO-SENSE) was developed for Smart India Hackathon 2026 under Problem Statement SIH26078, sponsored by the Ministry of Earth Sciences (MoES) and the National Centre for Medium Range Weather Forecasting (NCMRWF). The platform addresses the critical national challenge of disaster alert fatigue and untargeted mass warnings by transitioning India from broad 25-50km district-wide advisories to a precision 5km hyper-local danger zone polygon with coupled real-time evacuation and convoy logistics.

By fusing live multi-source station telemetry, NASA satellite imagery, GFS 30-member ensemble predictions, and OpenStreetMap road networks, VARUNA achieves a 65% to 85% reduction in unnecessary false-alarm areas, saving crucial administrative resources and preventing unnecessary economic paralysis during severe weather events.

2. COMPLETE WEBSITE NAVIGATION DIRECTORY & DIRECT ROUTES

The web application is fully deployed and accessible globally at https://varuna-access.vercel.app. Below is the complete architectural directory of all modules, their URLs, and operational functions:

Screen 1: Tactical Access Gateway

Direct Access URL: https://varuna-access.vercel.app/ or /#login

- Dual-Factor Authentication (Operator ID, Password, 6-digit Time-based OTP).
- Pre-authorized One-Click Test Credentials button for rapid demonstration.
- Live IST Synchronized Telemetry Clock and immutable cryptographic audit logging.

Screen 2: Live Synoptic Telemetry & Anomaly Tracker

Direct Access URL: https://varuna-access.vercel.app/#dashboard or /#tracker

- Real-time monitoring across 18 high-vulnerability national stations in 6 meteorological zones.
- Dual-API Cross-Verification: Live Open-Meteo telemetry cross-checked against NASA POWER MERRA-2.
- Automated Anomaly Detection: Auto-promotes stations to WATCH / WARNING when thresholds exceed sigma > 2.5.
- Neutral Radar Inspector: Clean default state on load; clicking any station unlocks deep analytical telemetry.

Screen 3: 4D Spatio-Temporal Track Reconstruction

Direct Access URL: https://varuna-access.vercel.app/#track or /#replay

- 7 Flagship Seeded Historical Disaster Benchmarks: Cyclone Amphan (2020), Cyclone Tauktae (2021), Mumbai Cloudburst (2005), Kedarnath Deluge (2013), Phalodi Heat Record (2016), Delhi Heatwave (2022), Churu Cold Wave (2019).
- 5 to 8 temporal steps per event with central coordinates, central pressure, wind speeds, and precipitation rates.
- Real NASA GIBS Satellite Sync: High-resolution MODIS Terra/Aqua True Color imagery dynamically matched to dates.
- Historical Pattern Comparison Card: Return periods, 30-year climatological normals, and top 3 analog matches.

Screen 4: Hyper-Local Alert Zone & IMD Bulletin Engine

Direct Access URL: https://varuna-access.vercel.app/#bulletin

- 5km Pinpoint Impact Polygon: Calculates actual geographic exposure and Census 2011 population affected.
- False-Alarm Area Reduction Gauge: Highlights 72% to 84% reduction versus conventional 50km radius.
- Automated IMD Standard Bulletin: Formats official 4-tier color-coded (Red/Orange/Yellow/Green) bilingual bulletins.
- Multilingual Public Broadcasts: Generates targeted citizen SMS and Audio alerts in English, Hindi, Bengali, Gujarati, Odia.

Screen 5: AI Predictor & Medium-Range Ensemble Outlook

Direct Access URL: https://varuna-access.vercel.app/#outlook

- 30-Member GFS Ensemble Processing: Computes Extreme Forecast Index (EFI) 1 to 5 days in advance.
- Probabilistic Risk Envelopes: Visualizes high-probability hazard corridors before convective genesis.
- Multi-Hazard Watch Zones: Proactive advisories across Western Ghats, Bay of Bengal, and Indo-Gangetic plains.

Screen 6: Evacuation & Relief Operations Command

Direct Access URL: https://varuna-access.vercel.app/#evacuation

- Tab 1 - Real Evacuation Routes: Live OpenStreetMap Overpass queries for real roads and designated disaster shelters. Performs live capacity vs. population matching, automatically flagging secondary evacuation if local shelters overflow.
- Tab 2 - Convoy Fleet Tracker: Real-time simulation of relief supply convoys (Food & Water, Medical Trauma Kits, Shelter Kits) navigating real road networks with live transit animation, speeds, and ETA countdowns.

Screen 7: Data Sources & Methodology Dossier

Direct Access URL: https://varuna-access.vercel.app/#methodology

- Comprehensive academic disclosure of all 9 integrated data APIs, update frequencies, and access types.
- Complete Mathematical Formulation of the 5km Pinpoint Radius and 4D Spatio-Temporal Tracking.
- What Is Real vs. What Is Modeled transparency disclosures and one-click PDF export.

Screen 8: Operational Security & System Settings

Direct Access URL: https://varuna-access.vercel.app/#settings

- API Quota & Latency Gauges: Displays live consumption for Open-Meteo, NASA POWER, and Overpass APIs.
- Cryptographic Audit Trail: Chronological log of every operator sign-in, alert dispatch, and configuration change.
- Role-Based Access Controls (RBAC): Enforces granular permissions for Analysts, Senior Meteorologists, and Directors.

Global Utility: Ask VARUNA (Tactical Circular FAB)

Direct Access URL: https://varuna-access.vercel.appAvailable on all operational screens

- Grounded Natural Language Assistant: Answers complex questions using live telemetry and active hazard states.
- Floating Action Button (FAB) design: High-tech pulsing tactical radar icon with zero UI obstruction.

3. INTEGRATED DATA SOURCES & GROUND-TRUTH MATRIX

A critical strength of Project VARUNA is its factual honesty and strict segregation between live real data, authoritative historical archives, and operational demonstration simulations. Below is the verified source matrix:

Data Source / API	Operational Application	Access Type	Refresh Cadence
Open-Meteo Live API	18-station surface telemetry & 30-year climate baselines	Public / No Key	~60s Polling
Open-Meteo Ensemble	30-member GFS ensemble for Extreme Forecast Index	Public / Open	Per Model Cycle
NASA POWER (MERRA-2)	Independent cross-verification of surface meteorology	Public / Open	Hourly Reanalysis
NASA GIBS Archive	True-color MODIS Terra/Aqua satellite imagery sync	Public Domain	Daily Satellite Passes
OpenStreetMap Overpass	Real road network geometries & verified shelter nodes	Open Database (ODbL)	Live Dynamic Query
NOAA IBTrACS v04	Ground-truth cyclone trajectories & central pressure	Public Domain	Archival Reference
Census of India 2011	District population densities for false-alarm calculations	Open Data Govt India	Official Decadal
IMD RSMC Bulletins	Official classification criteria & 4-tier alert protocols	Official Advisories	Procedural Baseline

4. STEP-BY-STEP RAPID EVALUATION GUIDE FOR JUDGES

RAPID DEMONSTRATION WORKFLOW FOR JUDGES & REVIEWERS

STEP 1: LOGIN & ACCESS CONTROL
- Navigate to https://varuna-access.vercel.app
- Click the green 'Use Authorized Test Credentials' button at the bottom of the login card.
- Click 'Authorize & Enter Mission Control' to access the live dashboard.

STEP 2: TELEMETRY & CROSS-VERIFICATION
- View the 18 live monitored stations. Observe dual-source agreement between Open-Meteo and NASA POWER.
- Click any station in the Anomaly Tracker to view real-time wind speed, surface pressure, and deviation sigma.

STEP 3: 4D TRACK RECONSTRUCTION & SATELLITE SYNC
- Navigate to 'Track & Timeline' in the sidebar.
- Select 'Cyclone Amphan' or 'Mumbai Extreme Cloudburst'.
- Drag the temporal scrubber or click 'Play 4D Track' to observe track progression synced with real NASA MODIS satellite imagery.

STEP 4: HYPER-LOCAL ALERT ZONE & BULLETINS
- Navigate to 'Alerts & Bulletin'. Select an event or What-If scenario.
- Observe the 5km pinpoint warning zone, the calculated false-alarm reduction percentage, and the auto-generated IMD bulletin.
- Test the multilingual audio/SMS broadcast generator in Hindi, Bengali, or Gujarati.

STEP 5: EVACUATION & CONVOY DISPATCH
- Navigate to 'EVACUATION & RELIEF OPS' in the sidebar.
- Under 'Evacuation Routes', inspect real roads and shelters queried via OSM Overpass API with live capacity matching.
- Under 'Convoy Tracker', watch animated relief convoys navigating real road networks toward designated shelters.